Check for updates

Received: 5 March 2020 | Revised: 3 July 2020 | Accepted: 5 July 2020

DOI: 10.1111/aas.13667

ORIGINAL ARTICLE

acta Anaesthesiologica Scandinavica

Comparing two airway management strategies for moderately sedated patients undergoing awake craniotomy: A single-blinded randomized controlled trial

Meng Deng[1] | Meng-Yun Tu[1] | Yi-Heng Liu[1] | Xiao-Bing Hu[1] | Tao Zhang[2] | Jin-Song Wu[3] | Ying-Wei Wang[1]

[1]Department of Anesthesiology, Huashan Hospital of Fudan University, Shanghai, China

[2]Department of Epidemiology, School of Public Health, Fudan University, Shanghai, China

[3]Department of Neurosurgery, Huashan Hospital of Fudan University, Shanghai, China

Correspondence
Ying-Wei Wang, Department of Anesthesiology, Huashan Hospital of Fudan University, Shanghai, China.
Email: wangyw@fudan.edu.cn

Funding information
This work was funded by the National Natural Science Foundation of China [grant numbers 81401089 to Meng Deng, 81730031 to Ying-Wei Wang], the Shanghai Municipal Key Clinical Specialty [grant numbers shslczdzk06901 to Ying-Wei Wang], and the National Key Technology R&D Program of China [grant number 2014BAI04B05 to Jin-Song Wu].

Abstract

Background: In the monitored anesthesia care (MAC) setting for awake craniotomy (AC), maintaining airway patency in sedated patients remains challenging. This randomized controlled trial aimed to compare the validity of the below-epiglottis transnasal tube insertion (the tip of the tube placed between the epiglottis and vocal cords) and the nasopharyngeal airway (simulated by the above-epiglottis transnasal tube with the tip of the tube placed between the epiglottis and the free edge of the soft palate) with respect to maintaining upper airway patency for moderately sedated patients undergoing AC.

Methods: Sixty patients scheduled for elective AC were randomized to receive below-epiglottis (n = 30) or above-epiglottis (n = 30) transnasal tube insertion before surgery. Moderate sedation was maintained in the pre- and post-awake phases. The primary outcome was the upper airway obstruction (UAO) remission rate (relieved obstructions after tube insertion/the total number of obstructions before tube insertion).

Results: The UAO remission rate was higher in the below-epiglottis group [100% (12/12) vs 45% (5/11); $P = .005$]. The tidal volume values monitored through the tube were greater in the below-epiglottis group during the pre-awake phase ($P < .001$). End-tidal carbon dioxide ($EtCO_2$) monitored through the tube was higher in the below-epiglottis group at bone flap removal ($P < .001$). During the awake phase, patients' ability to speak was not impeded. No patient had serious complications related to the tube.

Conclusion: The below-epiglottis tube insertion is a more effective method to maintain upper airway patency than the nasopharyngeal airway for moderately sedated patients undergoing AC.

Meng Deng, Meng-Yun Tu and Yi-Heng Liu contributed equally to this work.

© 2020 The Acta Anaesthesiologica Scandinavica Foundation. Published by John Wiley & Sons Ltd

1 | INTRODUCTION

For brain tumors in close proximity to eloquent areas, awake craniotomy (AC) has been proven to be a useful approach, enabling optimum resection while preserving functional integrity.[1,2] A variety of anesthetic techniques have been used for AC: the "asleep-awake-asleep" (AAA) technique in which the patient is under general anesthesia using a laryngeal mask airway (LMA) in the pre- and post-awake phases; the "asleep-awake" technique where the patient is kept awake from mapping to the end of surgery; the monitored anesthesia care (MAC) technique, which involves light to moderate sedation with spontaneous breathing; and the "fully awake" method with only local or regional anesthesia throughout the procedure.[3-5]

At our center, we prefer the MAC technique with moderate sedation because it provides comfort, does not require repeated airway interventions, and allows a rapid and smooth transition between the sedated and awake states. However, the lack of effective airway interventions in this setting presents a high risk of respiratory adverse events, such as upper airway obstruction (UAO) and ventilatory depression.[3,6,7]

UAO can occur at the level of the soft palate, tongue, and epiglottis.[8,9] The widely used nasopharyngeal airway, whose tip usually ends approximately 1 cm above the epiglottis,[9] can theoretically resolve UAO at the level of the soft palate and tongue but not at the level of the epiglottis. Therefore, we introduced an airway management strategy in which a transnasal tracheal tube was inserted with the tip slightly below the epiglottis and above the vocal cords to effectively resolve UAO at the three levels.

We hypothesized that compared to the above-epiglottis transnasal tube insertion (a tube tip position similar to that of the nasopharyngeal airway), the below-epiglottis transnasal tube insertion would safely improve the UAO remission rate in moderately sedated patients undergoing AC. The aim of this study was to determine the effectiveness and safety of this technique.

2 | MATERIALS AND METHODS

This prospective, randomized controlled trial was approved by the Institutional Review Board of Huashan Hospital of Fudan University (KY2017-382) on 23 November 2017 and registered at the Chinese Clinical Trial Registry (www.chictr.org.cn; identifier: ChiCTR-INR-17012411) on 18 August 2017. Written informed consent was obtained from all participants.

Between December 2017 and March 2019 at Huashan Hospital of Fudan University, patients aged 16 to 60 years with American Society of Anesthesiologists (ASA) physical status I-II, scheduled for elective AC for resection of supratentorial brain lesions were included. The exclusion criteria included patient refusal, claustrophobia, inability to remain still, cognitive disorders, mood instability, morbid obesity, nasopharyngeal airway placement contraindications (eg, nasal abnormalities and bleeding diatheses), and pregnancy. The criteria for unblinding included poor cooperation of the patient and unplanned serious medical conditions.

Editorial Comment

For awake craniotomy procedures where patients need to remain still, they can be treated with some sedating drugs for comfort, and nasopharyngeal tubes are commonly used to maintain a patent airway. This study aimed to identify an optimal tube tip position. The findings showed that a tube with the tip below the epiglottis is safe and reduces the prevalence of upper airway obstruction, compared to insertion with the tip above the epiglottis.

The Observer's Assessment of Alertness/Sedation (OAA/S) scale was used to determine the level of sedation. In line with previous studies,[10,11] we defined moderate sedation as 2 to 3 points on the OAA/S scale.

2.1 | Randomization and blinding

Eligible patients were randomly assigned to the below-epiglottis or above-epiglottis group at a 1:1 ratio using a computer-generated randomization table. The sealed opaque envelope technique was used for group allocation. A researcher not otherwise involved in the study opened the envelopes and placed the tube as indicated. Patients, neurosurgeons, and the outcome assessor were blinded to the technique used.

2.2 | Interventions

A transnasal tube (Polar Preformed Tracheal Tube; Smiths Medical Inc, Keene, NH, USA) was inserted under fiberoptic bronchoscopy guidance. All patients were moderately sedated under this procedure. The "spray-as-you-go" technique was used to apply 2% lidocaine when advancing the tube. In the below-epiglottis group, the tip of the tube was placed slightly below the epiglottis and approximately 1 to 1.5 cm above the vocal cords. In the above-epiglottis group, the tip of the tube was placed approximately 1 cm above the epiglottis.

2.3 | Anesthesia

On arrival in the operating room, electrocardiography, noninvasive blood pressure, and pulse oximetry (SpO_2) were applied. An arterial line was inserted under local anesthesia. Supplemental oxygen (4 L/min) was supplied through a regular nasal cannula. Sodium valproate (20-30 mg/kg), ondansetron (4 mg), and midazolam (1 mg) were administered intravenously. Dexmedetomidine infusion was started with 0.8-1 μg/kg over 15 min, followed by a maintenance infusion of 0.1-0.2 μg/kg/h. Target-controlled infusion (TCI) of propofol and remifentanil was initiated at 0.5 μg/

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

mL (Marsh model, plasma-targeting mode) and 0.5 ng/mL (Minto model, effect-site-targeting mode), respectively. Scalp nerve block was performed using 30-40 mL of 0.5% ropivacaine and 0.5% lidocaine with 1:200,000 epinephrine.[12]

After the loading dose of dexmedetomidine and the scalp nerve block were administered, the patient was connected to the anesthetic circuit via a tight but comfortably fitting anesthesia mask to measure tidal volume for the diagnosis of UAO. The target concentration of propofol was then increased by 0.1 μg/mL every 2 min to achieve moderate sedation. The occurrence of UAO was assessed by an outcome assessor. Then, the researcher entered the operating room and placed the transnasal tube according to the allocation. Other individuals in the operating room, including neurosurgeons and the outcome assessor, were blinded to this procedure. The tube was connected to the breathing circuit for supplemental oxygen (4 L/min) and respiration monitoring. Afterward, the outcome assessor continued to assess whether the pre-existing UAO was relieved. If UAO persisted, propofol or remifentanil infusion could be decreased or suspended at the discretion of the anesthesiologist responsible for anesthesia management. A chin lift or jaw thrust could be performed when necessary.

To facilitate intraoperative magnetic resonance imaging and surgical access, patients were kept in the supine position with the head rotated approximately 15°-60° contralateral to the tumor. A thin pillow could be placed under the shoulder to enhance the patient's positional endurance. The position of the tube tip was confirmed by the researcher after positioning. Mannitol 0.75 g/kg was administered at scalp incision to lower intracranial pressure (ICP). During craniotomy, the target effect-site concentration (Ce) of remifentanil could be increased to a maximum of 0.8 ng/mL, which permits spontaneous breathing,[13] to facilitate pain management as needed.

When the bone flap was removed, propofol infusion was discontinued and the infusion rates of dexmedetomidine and the target Ce of remifentanil were adjusted to 0.05-0.1 μg/kg/h and 0.5 ng/mL, respectively. Before dura opening, the neurosurgeons assessed the condition of brain relaxation through tactile evaluation of dural tension. The patient could be asked to breathe deeply, and the operating table could be tilted head up to improve brain relaxation if necessary. When brain mapping and tumor resection were completed, the infusion rates of propofol, remifentanil, and dexmedetomidine were resumed until the end of surgery. The tube was removed at head frame removal.

2.4 | Outcome variables

The primary outcome was the UAO remission rate (the ratio of relieved UAO cases after transnasal tube insertion to the UAO cases before transnasal tube insertion). According to previous studies,[14-16] we defined a UAO episode by a tidal volume measured through the anesthesia mask[17] lower than 200 mL over three successive breathing efforts accompanied by any clinical signs of continuing respiratory effort, including snoring, inspiratory stridor, or a combination of diminished chest wall movement and distortion with paradoxical inward motion of the anterior rib cage.[8,18] The UAO was considered relieved if the clinical signs disappeared after tube insertion.

Secondary outcomes included the tidal volume and end-tidal carbon dioxide ($EtCO_2$) monitored through the transnasal tube and patients' ability to speak during the awake phase. Tidal volume and $EtCO_2$ were recorded at four time points: T1, 5 min after transnasal tube insertion; T2, 5 min after placement of the head frame; T4, when removing the bone flap; and T3, the mean values of the tidal volume/$EtCO_2$ collected every 30 min from T2 to T4. During language mapping, if the patient's speaking was fluent and could be fully understood by the neurosurgeon, the ability to speak was considered unimpeded; otherwise, speech impediment was determined.

Other outcomes included the OAA/S scale, mean arterial pressure, and heart rate at T0 (5 min after arterial line establishment), T1, T2, T3, and T4; the Ce values of propofol and remifentanil, and the respiratory rate at T1, T2, T3, and T4; $PaCO_2$ at T4; complications related to the inserted tube; and intraoperative adverse events. Complications related to the inserted tube included sore throat, nasal congestion, cough, hoarseness, nausea, vomiting, laryngospasm, and tube blockage confirmed after removal. Intraoperative adverse events included oxygen desaturation, seizures, body movement, dry mouth, and shivering. At 24 h after surgery, patients were asked about their satisfaction with the operation.

2.5 | Statistical analysis

This study was powered to detect a difference in the UAO remission rate between the two groups, with the β value set at 0.2 and the α value set at 0.05 (two-sided). We estimated the sample size using PASS v11.0 (NCSS LLC., Kaysville, UT, USA). In a pilot study, the remission rates were 83% (5/6) in the below-epiglottis group and 25% (1/4) in the above-epiglottis group. Approximately 44% of moderately sedated patients were expected to develop UAO based on our unpublished observations. Therefore, a minimum of 25 cases were needed in each group. Considering a 20% dropout rate, we recruited 30 patients for each group.

Data were analyzed using SPSS software v22.0 (SPSS Inc, Chicago, IL, USA). Repeated measurements were analyzed with repeated-measures ANOVA and a Bonferroni-corrected Student's t-test for intergroup comparisons at each time point based on the normality test results. Other continuous variables were compared using the Mann-Whitney U test or Student's t-test, as appropriate. Categorical variables were compared using the χ^2 test or the Fisher's exact probability test, as appropriate. Ordinal data were compared between the groups using the Mann-Whitney U test. A two-sided $P < .05$ was considered significant unless Bonferroni adjustments (an adjusted significance level of .01) were performed.

3 | RESULTS

After screened for eligibility, 60 patients were equally randomized to the below-epiglottis group or the above-epiglottis

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

group (Figure 1). One patient in the above-epiglottis group developed persistent UAO under moderate sedation but refused to undergo craniotomy with light or no sedation before surgery. This patient was unblinded and treated with below-epiglottis tube insertion. Subsequently, the patient did not develop UAO under moderate sedation and completed the procedure. This patient was still analyzed in the above-epiglottis group based on the intention-to-treat principle. Patient characteristics and clinical characteristics were comparable between the two groups (Table 1).

3.1 | Primary outcome

The incidences of UAO were 40% (12/30) and 37% (11/30) in the below- and above-epiglottis groups ($P = .791$), respectively. After transnasal tube insertion, UAO was relieved in all 12 patients in the below-epiglottis group and in five patients in the above-epiglottis group. Thus, below-epiglottis tube insertion achieved a higher UAO remission rate than that was achieved in the above-epiglottis group [100% (12/12) vs 45% (5/11); difference, 55%; 95% CI, 19% to 79%; $P = .005$].

3.2 | Secondary outcomes

Repeated-measures ANOVA showed a significantly higher tidal volume monitored through the tube over time in the below-epiglottis group ($P < .001$; Figure 2A). The $EtCO_2$ monitored through the tube was higher in the below-epiglottis group at T4 ($P < .001$; Figure 2B). During the awake phase, no patient's ability to speak was impeded by the tube (Supplemental Material).

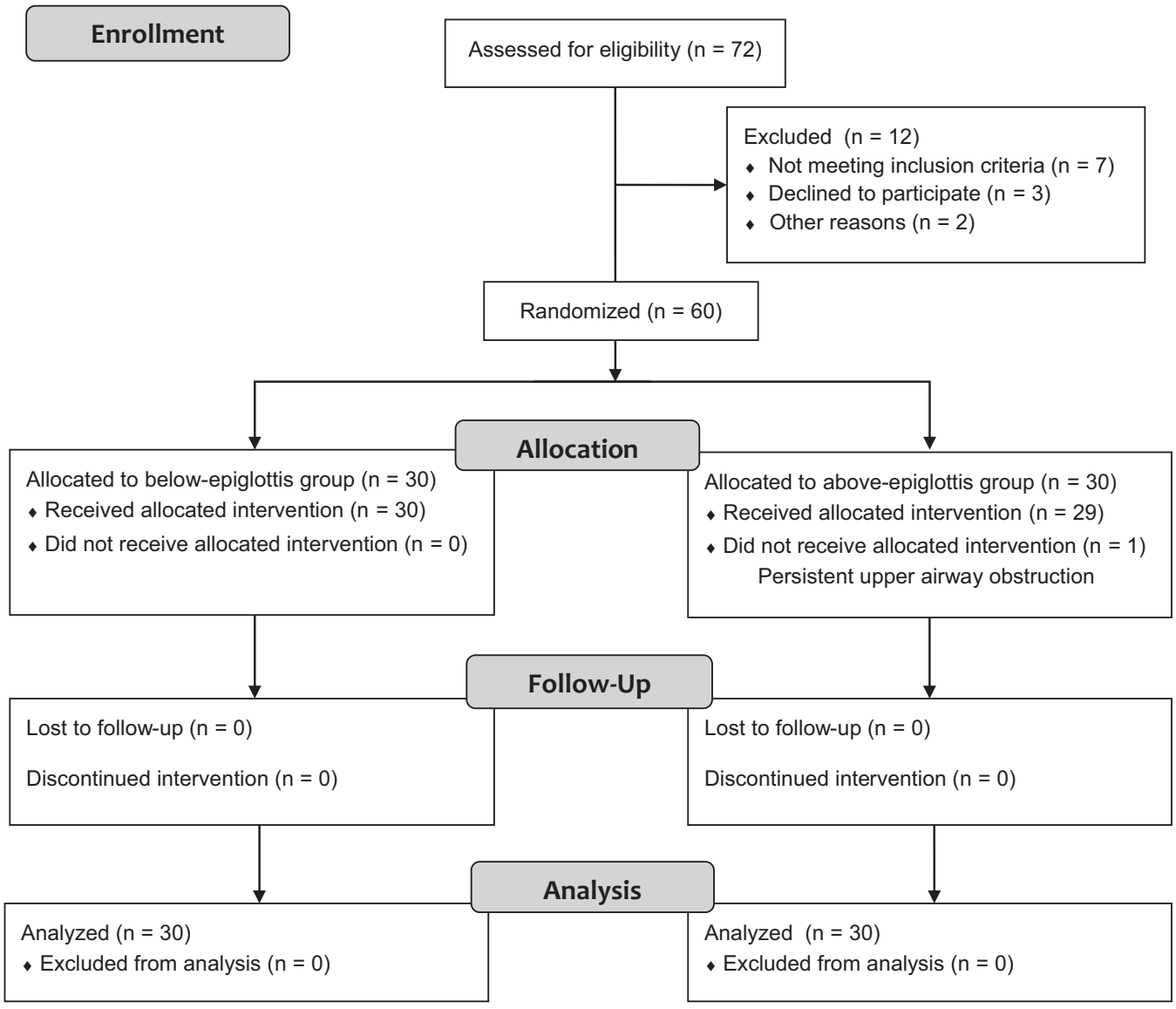


FIGURE 1 CONSORT flow diagram

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Patient characteristics and clinical characteristics

	Above-epiglottis Group (n = 30)	Below-epiglottis Group (n = 30)
Patient characteristics		
Age, yr, mean (SD)	37.4 (11.0)	36.0 (11.2)
Weight, kg, mean (SD)	61.0 (12.8)	63.0 (12.1)
Height, cm, mean (SD)	165.9 (7.9)	167.8 (7.4)
BMI, kg/m^2, mean (SD)	22.0 (3.4)	22.3 (3.4)
Sex: male/female, n (%)	13/17 (43/57)	17/13 (57/43)
ASA physical status: I/II, n (%)	2/28 (7/93)	1/29 (3/97)
Medical comorbidities, n (%)		
Pre-operative seizure	20 (67)	24 (80)
Snoring	7 (23)	6 (20)
Smoking	2 (7)	2 (7)
Rhinitis, pharyngitis	0 (0)	3 (10)
Upper respiratory tract infection in past 2 weeks	1 (3)	0 (0)
Airway assessment		
Mouth opening, cm, median [IQR]	4.5 [4.4-5.0]	4.5 [4.0-5.0]
Thyromental distance, cm, median [IQR]	6.5 [6.0-8.0]	6.5 [6.0-7.1]
Modified Mallampati classification, n (%)		
I	5 (17)	3 (10)
II	23 (77)	26 (87)
III	2 (7)	1 (3)
Arousal time, min, mean (SD)	10.6 (5.5)	12.2 (5.0)
Duration of awake phase, min, mean (SD)	142.4 (74.6)	134.0 (52.8)

Note: Data are presented as the mean (SD), number (proportion), or median [IQR].

Abbreviations: SD, standard deviation; IQR, interquartile range; ASA, American Society of Anesthesiologists; Arousal time, the time interval from discontinuation of propofol to the time when the patient could respond to verbal command.

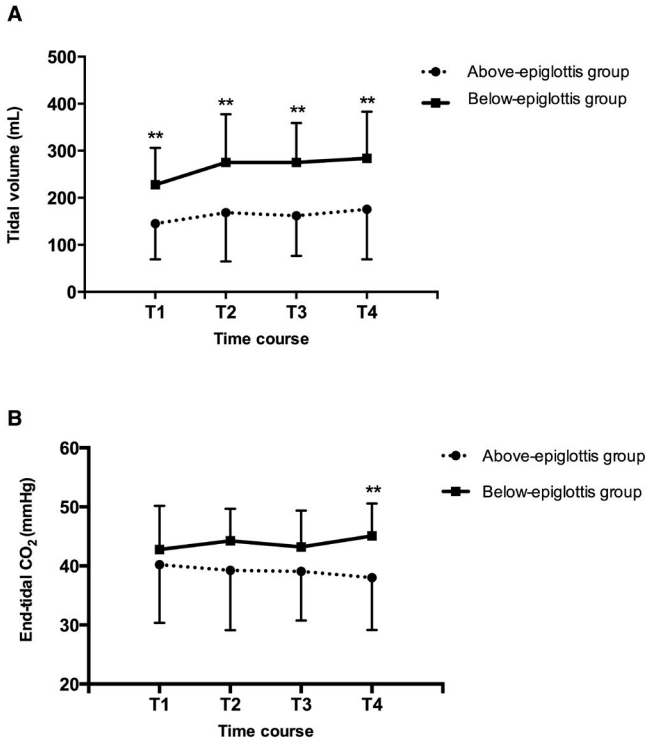


FIGURE 2 (A) Tidal volume and (B) end-tidal CO_2 monitored through the tube. The results are shown as the mean and standard deviation. In the below-epiglottis group, the tip of the tube was placed between the epiglottis and the vocal cords; in the above-epiglottis group, the tip of the tube was placed above the epiglottis. T1, 5 min after transnasal tube insertion; T2, 5 min after placement of the head frame; T4, when removing the bone flap; and T3, the mean value of the measurements collected every 30 min from T2 to T4. Significant *P* values were determined using a Bonferroni-adjusted α level (α = 0.01) for intergroup comparisons at individual time points. $^{**}P < .001$

3.3 | Other outcomes

The OAA/S scores were significantly lower in the below-epiglottis group at T3 ($P = .002$) and T4 ($P = .001$; Figure 3A). The Ce of propofol was higher in the below-epiglottis group at T3 ($P = .007$) and T4 ($P = .004$; Figure 3B). The Ce of remifentanil was not different between the groups ($P = .725$; Figure 3C). No difference was found between the below- and the above-epiglottis group in $PaCO_2$ at T4 [mean (standard deviation): 50.8 (4.7) vs 49.1 (5.1) mmHg; difference, 1.7 mmHg; 95% CI, −0.8 to 4.2; $P = .184$]. With regard to the respiratory rate, mean arterial pressure, and heart rate, no significant difference was found between the two groups (Figure 4).

No patient experienced cough, hoarseness, nausea, vomiting, laryngospasm, or oxygen desaturation throughout surgery. No significant difference was observed between the two groups in terms of complications related to the inserted tube, intraoperative adverse events (Table 2), or post-operative satisfaction ($P = 1.000$). Three patients had seizures during brain mapping and were successfully treated with cold saline irrigation on the brain surface and propofol boluses. Post-operatively, only one patient (a 16-year-old girl in the above-epiglottis group) complained of fear during the awake period and was not satisfied with the surgery.

4 | DISCUSSION

In this study, we introduced the below-epiglottis transnasal tube insertion for AC in our center and demonstrated that this technique was superior to the above-epiglottis tube insertion (nasopharyngeal airway) in resolving UAO for patients under moderate sedation. Furthermore, this technique did not affect patients' ability to speak during mapping or cause additional complications.

At this point, there is no consensus on the best anesthetic technique for AC. The AAA setting provides airway protection and keeps

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

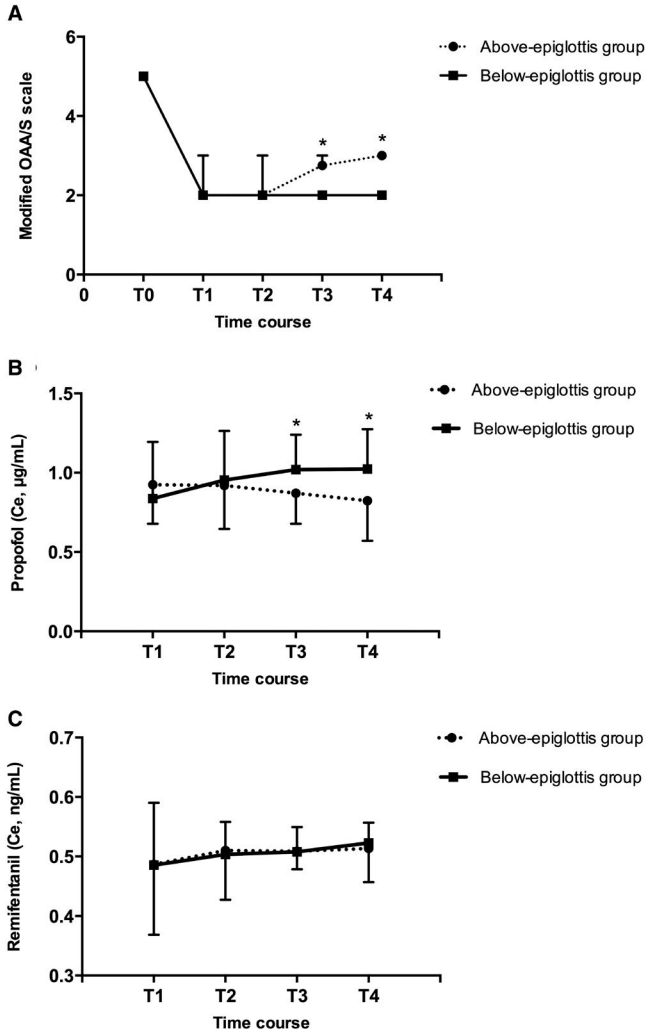


FIGURE 3 (A) Modified Observer's Assessment of Alertness/Sedation (OAA/S) scale; (B) the effect-site concentration (Ce) of propofol; and (C) the Ce of remifentanil. The OAA/S scale is shown as the median and interquartile range, and the Ce values of propofol and remifentanil are shown as the mean and standard deviation. In the below-epiglottis group, the tip of the tube was placed between the epiglottis and the vocal cords; in the above-epiglottis group, the tip of the tube was placed above the epiglottis. T0, 5 min after arterial line establishment; T1, 5 min after transnasal tube insertion; T2, 5 min after placement of the head frame; T4, when removing the bone flap; and T3, the mean value of the measurements collected every 30 min from T2 to T4. Significant *P* values were determined using a Bonferroni-adjusted α level (α = 0.01) for intergroup comparisons at individual time points. *$P < .01$

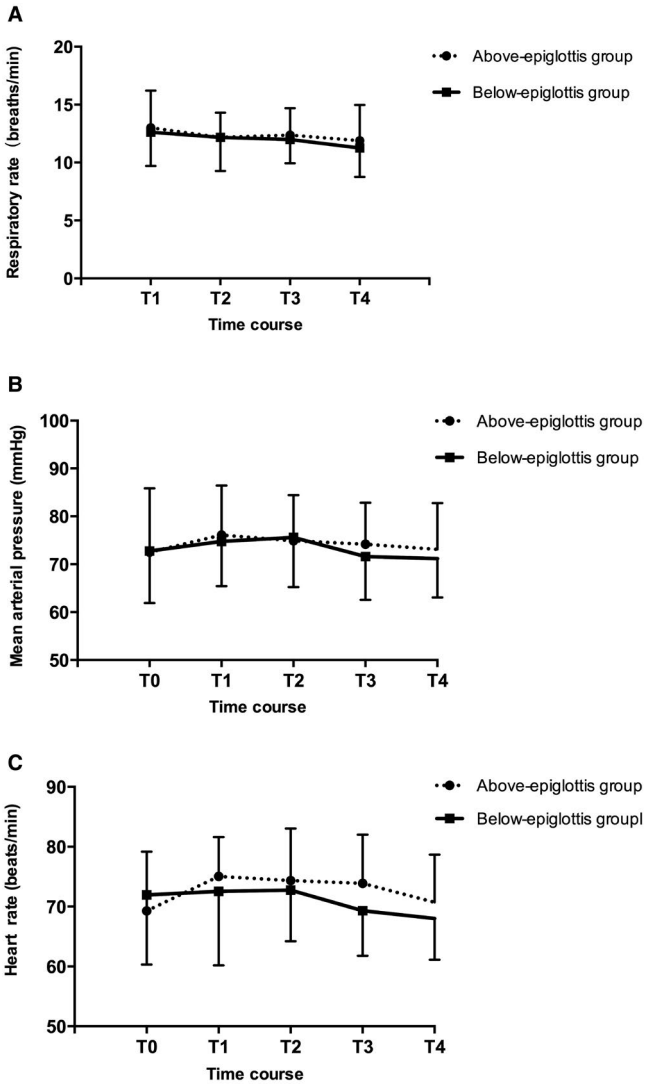


FIGURE 4 (A) Respiratory rate; (B) mean arterial pressure; and (C) heart rate. The results are shown as the mean and standard deviation. T0, 5 min after arterial line establishment; T1, 5 min after transnasal tube insertion; T2, 5 min after placement of the head frame; T4, when removing the bone flap; and T3, the mean value of the measurements collected every 30 min from T2 to T4

the patient motionless. However, this strategy requires removal and repositioning of the LMA intraoperatively, which might cause sudden cough, nausea, and elevated ICP.[17] Moreover, the unpredictable asleep-to-awake transition may unnecessarily prolong the procedure. When head rotation is required, an LMA cannot guarantee satisfactory controlled ventilation, which may result in air leak and ineffective ventilation.[4,5]

On the other hand, without an effective airway management, patients are exposed to potential hazards, such as UAO and ventilatory depression, in the MAC setting.[3] Such episodes occur because all available anesthetics, including propofol, remifentanil, and even dexmedetomidine, which are believed to have a low respiratory depressant effect, induce changes in the muscle tone of the upper airway and the central respiratory drive in a dose-dependent manner.[19-21]

4.1 | The below-epiglottis transnasal tube insertion can effectively resolve UAO

Unilateral or bilateral nasopharyngeal airway has been used in AC when spontaneous breathing is maintained.[22] However, theoretically, the nasopharyngeal airway cannot relieve an obstruction caused by the collapse of the epiglottis because its tip usually ends above the epiglottis. Accordingly, as demonstrated by our results, after tube insertion, UAO still occurred in six patients in the above-epiglottis

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Complications related to the inserted tube and intraoperative adverse events

	Above-epiglottis Group (n = 30)	Below-epiglottis Group (n = 30)	*P* value
Complications related to the inserted tube			
Sore throat	3 (10)	3 (10)	1.000
Nasal congestion	5 (17)	2 (7)	0.421
Tube blockage	5 (17)	1 (3)	0.197
Cough, hoarseness, nausea, and vomiting, laryngospasm	0 (0)	0 (0)	—
Intraoperative adverse events			
Oxygen desaturation	0 (0)	0 (0)	—
Seizure	2 (7)	1 (3)	1.000
Body movement	2 (7)	0 (0)	0.492
Dry mouth	2 (7)	1 (3)	1.000
Shivering	0 (0)	1 (3)	1.000

Note: Data are expressed as number (proportion). The χ^2 tests or Fisher's exact probability test were used as appropriate. $P < .05$ was used to denote statistical significance.

group. By contrast, the UAO remission rate was 100% (12/12) in the below-epiglottis group, suggesting that the below-epiglottis tube insertion prevented the collapse of all supraglottic soft tissue, including the soft palate, tongue, and epiglottis. As demonstrated by lower OAA/S scores and higher Ce values of propofol in the below-epiglottis group, this effective approach allowed moderate sedation to be safely applied during the pre-awake phase of AC.

4.2 | Patients' tolerance and safety evaluation of the below-epiglottis tube insertion method

To conduct functional mapping during the awake phase, the applied airway intervention should not impede patients' ability to speak. In this study, all patients could tolerate the tube whether they were moderately sedated or awake. Topical application of lidocaine to larynx structures and continuous infusion of anesthetics might have helped patients to tolerate the tube. In addition, their speaking was fluent and clear. These findings are consistent with those of Soehle M. et al,[13,23] who applied a similar airway management strategy in AC. They also stated that the tube could be inserted into the trachea under fiberoptic guidance if necessary. In that case, the below-epiglottis transnasal tube insertion would be a time-saving method to secure the airway and provide ventilatory support when unplanned serious situations occur.

A key question regarding this technique is whether it causes additional complications or adverse events. In theory, complete closure of the epiglottis is impeded by the tube and patients are at the risk of aspiration caused by vomiting. However, more important, the closure of vocal cords is intact.[24] Laryngospasm could also be a concern. However, we did not encounter any case of this, probably due to the anesthesia of the laryngeal mucosa and the 1-cm distance from the tip of the tube to the vocal cords. This distance probably remained constant as the tube was relatively fixed in the nasal cavity. Therefore, direct stimulation of larynx structures was avoided. As for other complications and adverse events, the incidences were not significantly different between the two groups. Accordingly, we found that compared to the above-epiglottis, the below-epiglottis transnasal tube insertion is a safe and more useful method for airway maintenance in AC.

Unexpectedly, the $PaCO_2$ when removing the bone flap was slightly higher than normal, which might have resulted from the respiratory depressant effect of anesthetics. In our hospital, patients could be asked to breathe deeply, and a head-up posture could be applied at the discretion of the neurosurgeon before dura opening. Mannitol was also applied to decrease ICP.[25] By doing so, the clinical conditions were acceptable in most cases. Other scholars also found that the AC proceeded uneventfully despite the $PaCO_2$ reaching the upper limit of or slightly exceeding the normal range.[5,20,26,27] A possible explanation is that brain relaxation could be affected by multiple factors. The influence of slightly increased $PaCO_2$ might be minor if other measures are jointly and appropriately adopted. Additionally, spontaneous breathing facilitates venous return,[28] thereby reducing the impact of increased $PaCO_2$ on brain relaxation during AC to some extent. However, this result suggests that timely blood gas analysis, especially before dura opening, should be performed to ensure that the patient has recovered adequate ventilation with a $PaCO_2$ in the normal range.

4.3 | Study limitations

There are several limitations to this clinical trial. First, after transnasal tube insertion, we did not measure the actual tidal volume to diagnose UAO remission. Second, using the subjective OAA/S scale combined with objective parameters, such as the bispectral index, would be better to assess the level of sedation. Third, the sample size, which was determined by the UAO remission rate, might be too small to detect a difference in secondary and other outcomes. Fourth, we did not record the condition of brain relaxation or the

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$PaCO_2$ at dura opening to explain the impact of mild hypercapnia at T4 on surgery. Based on our experience and other studies,[5,20,26,27] mild hypercapnia seems to be permissible in AC.

In conclusion, this randomized controlled clinical trial demonstrates that the below-epiglottis transnasal tube insertion was superior to the nasopharyngeal airway in maintaining upper airway patency and can be safely applied in moderately sedated patients undergoing AC.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

ORCID

Meng Deng https://orcid.org/0000-0002-1484-3039
Yi-Heng Liu https://orcid.org/0000-0002-7778-2368
Ying-Wei Wang https://orcid.org/0000-0002-2554-6611

REFERENCES

1. Serletis D, Bernstein M. Prospective study of awake craniotomy used routinely and nonselectively for supratentorial tumors. *J Neurosurg*. 2007;107:1-6.
2. Brown T, Shah AH, Bregy A, et al. Awake craniotomy for brain tumor resection: the rule rather than the exception? *J Neurosurg Anesthesiol*. 2013;25:240-247.
3. Goettel N, Bharadwaj S, Venkatraghavan L, Mehta J, Bernstein M, Manninen PH. Dexmedetomidine vs propofol-remifentanil conscious sedation for awake craniotomy: a prospective randomized controlled trial. *Br J Anaesth*. 2016;116:811-821.
4. Meng L, McDonagh DL, Berger MS, Gelb AW. Anesthesia for awake craniotomy: a how-to guide for the occasional practitioner. *Can J Anaesth*. 2017;64:517-529.
5. Olsen KS. The asleep-awake technique using propofol-remifentanil anaesthesia for awake craniotomy for cerebral tumours. *Eur J Anaesthesiol*. 2008;25:662-669.
6. Manninen PH, Balki M, Lukitto K, Bernstein M. Patient satisfaction with awake craniotomy for tumor surgery: a comparison of remifentanil and fentanyl in conjunction with propofol. *Anesth Analg*. 2006;102:237-242.
7. Elbakry AE, Ibrahim E. Propofol-dexmedetomidine versus propofol-remifentanil conscious sedation for awake craniotomy during epilepsy surgery. *Minerva Anestesiol*. 2017;83:1248-1254.
8. Hillman DR, Platt PR, Eastwood PR. The upper airway during anaesthesia. *Br J Anaesth*. 2003;91:31-39.
9. Stoneham MD. The nasopharyngeal airway Assessment of position by fibreoptic laryngoscopy. *Anaesthesia*. 1993;48:575-580.
10. Chernik DA, Gillings D, Laine H, et al. Validity and reliability of the observer's assessment of alertness/sedation scale: study with intravenous midazolam. *J Clin Psychopharmacol*. 1990;10:244-251.
11. Xi C, Sun S, Pan C, Ji F, Cui X, Li T. Different effects of propofol and dexmedetomidine sedation on electroencephalogram patterns: wakefulness, moderate sedation, deep sedation and recovery. *PLoS One*. 2018;13(6):e0199120. https://doi.org/10.1371/journal.pone.0199120
12. Bebawy JF, Bilotta F, Koht A. A modified technique for auriculotemporal nerve blockade when performing selective scalp nerve block for craniotomy. *J Neurosurg Anesthesiol*. 2014;26:271-272.
13. Soehle M, Wolf CF, Priston MJ, et al. Propofol pharmacodynamics and bispectral index during key moments of awake craniotomy. *J Neurosurg Anesthesiol*. 2018;30:32-38.
14. Mathru M, Esch O, Lang J, et al. Magnetic resonance imaging of the upper airway: effects of propofol anesthesia and nasal continuous positive airway pressure in humans. *Anesthesiology*. 1996;84:273-279.
15. Reber A, Wetzel SG, Schnabel K, Bongartz G, Frei FJ. Effect of combined mouth closure and chin lift on upper airway dimensions during routine magnetic resonance imaging in pediatric patients sedated with propofol. *Anesthesiology*. 1999;90:1617-1623.
16. The American Academy of Sleep Medicine Task Force. Sleep-related breathing disorders in adults: recommendations for syndrome definition and measurement techniques in clinical research. The report of an American academy of sleep medicine task force. *Sleep*. 1999;22:667-689.
17. Doufas AG, Shafer SL, Rashid NHA, Kushida CA, Capasso R. Nonsteady state modeling of the ventilatory depressant effect of remifentanil in awake patients experiencing moderate-to-severe obstructive sleep apnea. *Anesthesiology*. 2019;130:213-226.
18. Boriosi JP, Zhao Q, Preston A, Hollman GA. The utility of the pretracheal stethoscope in detecting ventilatory abnormalities during propofol sedation in children. *Paediatr Anaesth*. 2019;29(6):604-610.
19. King AB, Alvis BD, Hester D, Taylor S, Higgins M. Randomized trial of a novel double lumen nasopharyngeal catheter versus traditional nasal cannula during total intravenous anesthesia for gastrointestinal procedures. *J Clin Anesth*. 2017;38:52-56.
20. Sarang A, Dinsmore J. Anaesthesia for awake craniotomy–evolution of a technique that facilitates awake neurological testing. *Br J Anaesth*. 2003;90:161-165.
21. Lodenius A, Maddison KJ, Lawther BK, et al. Upper airway collapsibility during dexmedetomidine and propofol sedation in healthy volunteers: a nonblinded randomized crossover study. *Anesthesiology*. 2019;131:962-973.
22. Sivasankar C, Schlichter RA, Baranov D, Kofke WA. Awake craniotomy: a new airway approach. *Anesth Analg*. 2016;122:509-511.
23. Soehle M, Wolf CF, Priston MJ, et al. Comparison of propofol pharmacokinetic and pharmacodynamic models for awake craniotomy: a prospective observational study. *Eur J Anaesthesiol*. 2015;32:527-534.
24. Sudarshan R, Jadcherla WJH, Shaker R. Physiology and pathophysiology of glottic reflexes and pulmonary aspiration: from neonates to adults. *Semin Respir Crit Care Med*. 2010;31:554-560.
25. Orešković D, Maraković J, Varda R, Radoš M, Jurjević I, Klarica M. New insight into the mechanism of mannitol effects on cerebrospinal fluid pressure decrease and craniospinal fluid redistribution. *Neuroscience*. 2018;392:164-171.
26. Dilmen OK, Akcil EF, Oguz A, Vehid H, Tunali Y. Comparison of conscious sedation and asleep-awake-asleep techniques for awake craniotomy. *J Clin Neurosci*. 2017;35:30-34.
27. Rajan S, Cata JP, Nada E, Weil R, Pal R, Avitsian R. Asleep-awake-asleep craniotomy: a comparison with general anesthesia for resection of supratentorial tumors. *J Clin Neurosci*. 2013;20:1068-1073.
28. Soni N, Williams P. Positive pressure ventilation: what is the real cost? *Br J Anaesth*. 2008;101:446-457.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: Deng M, Tu M-Y, Liu Y-H, et al. Comparing two airway management strategies for moderately sedated patients undergoing awake craniotomy: A single-blinded randomized controlled trial. *Acta Anaesthesiol Scand*. 2020;64:1414–1421. https://doi.org/10.1111/aas.13667

13996576, 2020, 10, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/aas.13667 by Hacettepe University, Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License